

ManufactureERP

Software Requirements Specification — Extended Edition

Chapter 21 — PDF Generation Specifications
Chapter 22 — Live End-to-End Use Cases

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Extends SRS v1.0 — On-Premise Laravel ERP for Indian Manufacturing SMEs

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PDF Generation Specifications

ManufactureERP uses DomPDF (via the laravel-dompdf package) as its primary PDF rendering engine. Every document type has a dedicated Blade template, a Laravel PDF job class, and a structured delivery pipeline. This chapter specifies the exact content, layout, data sources, and generation logic for every PDF in the system.

21.1 PDF Engine Architecture

Technology: barryvdh/laravel-dompdf v2.x | **Queue:** All PDF jobs pushed to pdf-generation queue via Laravel Horizon | **Storage:** Generated PDFs stored in storage/app/documents/{module}/{year-month}/ | **Delivery:** Download link via signed URL (expires 24h) or direct WhatsApp attachment

1 Controller / Event Triggers PDF

Route or event calls PdfGeneratorService::generate(\$model). A dedicated job class is dispatched to the queue.

2 Blade Template Rendered

The queued job loads the appropriate Blade view (resources/views/pdfs/{module}/{template}.blade.php) and passes the Eloquent model with eager-loaded relations.

3 DomPDF Renders HTML → PDF

Dompdf::loadView() converts the HTML+CSS to PDF. Paper size A4, orientation portrait (landscape for reports). Font: DejaVu Sans (Unicode support for ■ symbol).

4 PDF Stored & URL Generated

PDF saved with hashed filename. Signed URL generated via URL::temporarySignedRoute() expiring in 24 hours. URL stored on the model record.

5 Delivery

User downloads via browser. OR job fires WhatsApp notification attaching the PDF URL. OR email via Laravel Mail with PDF as attachment.

Global PDF Header/Footer Blade Component:

All PDFs share a master layout (layouts/pdf-master.blade.php) that injects: company logo (top-left), company name + GSTIN + address (top-right), document title centred below header, and footer with page number + 'This is a computer-generated document.' + generation timestamp.

21.2 GST Tax Invoice PDF

The most critical document. Generated on invoice POST. Required for e-Invoice (IRN), customer dispatch, and GST filing. Must comply with GST Rule 46.

Section	Content / Data Source	Notes
Header Band	Company name, GSTIN, address, logo — from companies table	Mandatory GST Rule 46
Document Title	'TAX INVOICE' in bold caps centred	Rule 46 mandatory label
Invoice Meta	Invoice No, Invoice Date, Due Date, Place of Supply, Reverse Charge Y/N	All from invoices table

Section	Content / Data Source	Notes
Customer Block	Bill-to: company_name, GSTIN, address, state code. Ship-to if different	customers + invoice fields
Line Items Table	Sr, Item, HSN/SAC, Qty, UOM, Unit Price, Discount, Taxable Amt	invoice_items joined items
GST Breakup	Per HSN row: CGST Rate+Amt, SGST Rate+Amt OR IGST Rate+Amt	Computed from tax_type
Totals Block	Subtotal, Discount, Taxable Total, CGST, SGST/IGST, TCS, Round-off, Grand Total	All system-calculated
Amount in Words	■ [Grand Total] Only — auto-generated via NumberToWords helper	invoices.amount_in_words
Bank Details	Company bank name, account no, IFSC for NEFT payment	companies.bank_* fields
e-Invoice Block	IRN (64-char hash), Ack No, Ack Date, QR Code image	Only if einvoice_enabled
e-Way Bill	EWB number if applicable (goods > 50,000 or inter-state)	invoices.eway_bill_number
Signature Block	'For [Company Name]' + 'Authorised Signatory'	Static text
Footer	T&C;, page number, 'Computer generated invoice'	From company settings

■ **e-Invoice Rule:** If company.einvoice_enabled = true AND invoice is B2B, the system MUST obtain IRN from the IRP (Invoice Registration Portal) API BEFORE generating the final PDF. The QR code returned by IRP is embedded as a base64 image in the PDF. Draft PDFs without IRN are watermarked 'NOT VALIDATED'.

21.3 Purchase Order PDF

Generated when PO status moves to APPROVED or SENT. Sent to vendor via email + WhatsApp.

Section	Content / Data Source
Header	Company name, address, GSTIN, logo
Document Title	'PURCHASE ORDER' centred
PO Meta	PO Number, PO Date, Expected Delivery Date, Payment Terms
Vendor Block	Vendor company name, contact, GSTIN, address — from vendors table
Delivery To	Warehouse name + address — from warehouses table
Line Items	Sr, Item Code, Description, HSN, Qty, UOM, Unit Price, GST%, Total
Tax Summary	CGST/SGST or IGST breakdown same as invoice
Totals	Subtotal, Tax, Grand Total, Amount in Words
Terms Block	Payment terms, delivery terms, quality standards (free text from PO notes)

Section	Content / Data Source
Approval Block	Approved by (name + designation), approval date, digital stamp placeholder
Footer	'This PO is subject to terms and conditions on reverse.' Page no.

21.4 Goods Receipt Note (GRN) PDF

Internal document. Generated after GRN is POSTED. Used for 3-way matching and QC record.

Section	Content
Header	Company logo, name, GRN number, date, warehouse name
Reference Block	Linked PO Number, Vendor name, Vehicle No, LR Number
Items Received	Item code, name, PO qty, Received qty, Accepted qty, Rejected qty, Batch/Serial, Expiry Date
QC Section	QC status per line (PASS/FAIL/PARTIAL), QC remarks, QC officer name
Variance	Any qty discrepancy highlighted in red
Posting Block	Posted by (user), posted at timestamp — immutable record proof
Footer	GRN number + barcode (code128) for scanning

21.5 Quotation PDF

Section	Content
Header	Company details, 'QUOTATION' title, Quotation No, Date, Valid Until date
Customer	Company name, address, contact person, GSTIN
Line Items	Sr, Product, Description, HSN, Qty, UOM, Unit Price, Discount%, GST%, Amount
Tax Summary	GST breakup (indicative — final tax calculated on invoice)
Totals	Subtotal, Discount, Tax, Total — with 'Indicative pricing' watermark note
Terms	Payment terms, validity, delivery timeline, special conditions from quotation.notes
Footer	'Valid until [date]. Subject to availability. Prices exclusive of freight unless stated.'

21.6 Delivery Challan PDF

Accompanies goods during transit. Required for e-Way Bill. Generated at dispatch step.

Section	Content
Header	Company name, GSTIN, address, 'DELIVERY CHALLAN' title
Challan Meta	Challan No, Date, Related Sales Order No, Related Invoice No
From-To	Ship from: Warehouse address. Ship to: Customer shipping address

Section	Content
Transport	Transporter name, vehicle number, LR/docket number, e-Way Bill number
Items	Item, Description, HSN, Qty, UOM — NO prices (challan is not a tax document)
Declaration	'Goods dispatched for delivery only. No commercial transaction.'
Signature	Driver signature, receiver signature (blank box), company stamp

21.7 Payslip / Salary Slip PDF

Generated for each employee during payroll run. Delivered via WhatsApp (WA-12). Password-protected with employee's DOB (DDMMYYYY) if encryption enabled.

Section	Content / Data Source	Notes
Header	Company name, logo, 'SALARY SLIP' — Month & Year prominent	
Employee Block	Employee Code, Name, Designation, Department, Date of Joining, Bank A/c ****last4	employees table
Attendance Box	Working days in month, Present, Absent, LOP (Loss of Pay) days, Leave taken (type-wise)	attendance + leave_applications
Earnings Table	Basic Salary, HRA, Conveyance, Special Allowance, Overtime, Other Allowances — each row with amount	payroll_details
Deductions Table	PF Employee (12%), ESI Employee (0.75%), Professional Tax, TDS, LOP deduction, Advance recovery, Other — each with amount	payroll_details
Employer Share	PF Employer (12%), ESI Employer (3.25%) — shown for reference, NOT deducted from salary	payroll_details
Net Pay Block	Gross Earnings, Total Deductions, NET PAY in box — prominent font	payroll_details.net_salary
Amount Words	Net Pay in words	Auto-generated
PF Details	UAN number, PF number, EPF wages, EPS wages — for PF reporting	employees.pf_number + uan
ESI Details	ESI number, ESI wages — only if applicable (gross <= 21,000)	employees.esi_number
Footer	'This is a computer-generated payslip. No signature required.' + company seal	

PF Calculation Logic: PF applies only if employee.basic_salary <= ₹15,000. Employee contribution = 12% of basic (max ₹1,800). Employer contribution = 12% of basic (split: 8.33% to EPS, 3.67% to EPF). If basic > ₹15,000, PF is optional — flag stored in employees.pf_opted_in.

ESI Calculation Logic: ESI applies if gross salary <= ₹21,000. Employee = 0.75% of gross. Employer = 3.25% of gross. Calculated on actual gross including allowances.

21.8 Payroll Summary Report PDF

Management-level report per payroll run. One row per employee, summary totals at bottom.

Column	Description
Emp Code + Name	From employees table
Department	From departments table
Working Days	Calendar days – holidays – LOP
Present Days	From attendance table
LOP Days	Loss of Pay days this month
Gross Salary	Sum of all earnings
PF (EE)	Employee PF deduction
ESI (EE)	Employee ESI deduction
Prof. Tax	State professional tax slab
TDS	Income tax deducted at source
Other Deductions	Advances, loan EMIs, etc.
Total Deductions	Sum of all deductions
Net Salary	Gross – Total Deductions
Bank A/c	Last 4 digits + IFSC for payment
Payment Status	PENDING / PAID / HOLD

The final row shows department-wise and company-total aggregates. A separate statutory summary box shows total PF payable to EPFO and total ESI payable to ESIC.

21.9 Stock Ledger Report PDF

Item-wise, date-range filtered report showing all stock movements. Running balance column enables audit trail verification.

Column	Description
Date	Transaction date
Transaction Type	GRN_IN / SALES_OUT / PRODUCTION_OUT / TRANSFER_IN etc.
Reference	Source document number (PO/GRN/SO/PRD)
Party	Vendor / Customer / Internal depending on type
Batch / Serial	If batch/serial tracked
Qty In	Stock added (positive movement)
Qty Out	Stock removed (negative movement)
Balance Qty	Running balance — critical for audit

Column	Description
Unit Cost	Cost at time of transaction (weighted average)
Value	Qty × Unit Cost

21.10 GSTR-1 / GSTR-3B Export PDF

Month-end compliance reports. The actual filing is done on the GST portal via JSON export, but the PDF serves as a printed record and review copy.

Report Section	GSTR-1 Content	GSTR-3B Content
B2B Invoices	All GST invoices to registered buyers (GSTIN-wise)	Not applicable
B2C Large	Inter-state invoices to unregistered (>₹2.5L)	Not applicable
B2C Small	Intra-state + small inter-state unregistered	Not applicable
CDN	Credit/Debit Notes issued	Included in net figures
Tax Summary	HSN-wise summary with CGST/SGST/IGST	Tax liability summary
Input Tax Credit	Not in GSTR-1	ITC claimed (purchase invoices)
Net Tax Payable	Not in GSTR-1	Output Tax – ITC = Payable
Nil-rated/Exempt	Nil + exempt supply summary	Included

21.11 Vendor Statement PDF

Account statement sent to vendor for reconciliation. Shows all POs, GRNs, invoices, and payments.

Column	Description
Date	Document date
Document Type	PO / GRN / Invoice / Payment / Debit Note / Credit Note
Reference No	Document number
Debit (Dr)	Amount owed to vendor (GRN / invoice posted)
Credit (Cr)	Amount paid / credit notes
Balance	Running balance — positive = payable to vendor

21.12 Production Order PDF

Shop floor document. Printed and given to production operators with BOM and instruction.

Section	Content
Header	Company name, 'PRODUCTION ORDER', Order No, Date
Product Block	Finished good name, item code, planned qty, UOM, target warehouse
Schedule	Planned start date, planned end date, batch number (if batch-tracked)
BOM Table	Sr, Raw Material code, name, Required Qty per unit, Total Qty for this order, Current Stock, Status (OK / SHORT)
Special Instructions	Free text from production_orders.notes
Sign-off Block	Supervisor name, Released by, Completed by — all with date/time fields
QC Checklist	Pre-defined quality checkpoints with Pass/Fail checkboxes

21.13 PDF Security & Delivery

Aspect	Specification
Watermark	Drafts and cancelled documents watermarked 'DRAFT' or 'CANCELLED' diagonally in red
Password Protection	Payslips optionally encrypted with employee DOB (DDMMYYYY) as password
Signed URLs	All download links are Laravel signed routes with 24-hour expiry
Storage	storage/app/documents/{module}/{YYYY-MM}/{uuid}.pdf — served via StreamedResponse
Audit Trail	PDF generation event logged in activity_log with user, timestamp, document reference
Regeneration	PDFs can be regenerated (new file, old URL invalidated). History tracked.
WhatsApp Delivery	PDF URL (not file) sent via WhatsApp template with signed URL. Meta stores link only.
Email Attachment	Invoices and payslips attached as PDF to outgoing emails via Laravel Mail

Live End-to-End Use Cases

This chapter walks through complete, realistic scenarios as they happen day-to-day in an Indian manufacturing company. Each use case names the actors, the trigger, every system step, the documents generated, and the notifications fired — giving the engineering and QA team a concrete script for development and testing.

22.1 Use Case: Full Procurement Cycle — Pharma Raw Material

Company	Suncure Pharma Pvt. Ltd., Ahmedabad (Gujarat — State 24)
Industry	Pharmaceuticals
Actors	Ravi (Store Keeper), Priya (Purchase Manager), Mehul (Accountant), VitroChem Suppliers (Vendor)
Trigger	Paracetamol API stock falls below reorder level of 50 kg

1 System Fires Low-Stock Alert

Laravel Scheduler runs CheckReorderLevels job at 8:00 AM daily. Finds Paracetamol API (item_code: RM-0042) stock = 38 kg < reorder_level = 50 kg. WhatsApp WA-03 fired to Priya's number: 'LOW STOCK: Paracetamol API — 38 kg remaining. Reorder qty: 200 kg.'

2 Ravi Raises Purchase Requisition

Ravi logs in, navigates to Purchase → New Requisition. Selects item RM-0042, qty 200 kg, required date 3 days from today, notes 'Urgent — production scheduled for next week'. PR/2026-04/0023 created. Status: PENDING_APPROVAL.

3 Priya Approves PR

Priya receives system notification. Reviews PR. Checks current price history. Clicks Approve. PR status → APPROVED. Ravi's WhatsApp: 'Your PR/2026-04/0023 has been approved.'

4 Priya Creates Purchase Order

Priya clicks 'Convert to PO' from approved PR. Vendor VitroChem Suppliers auto-selected (preferred vendor for RM-0042). Unit price ₹420/kg, GST 18% (CGST 9% + SGST 9% — intra-state). PO total = 200 kg × ₹420 + ₹15,120 GST = ₹99,120. PO/2026-04/0018 created. Status: DRAFT.

5 PO Approval Workflow

PO amount ₹99,120 > approval threshold ₹50,000 (configured). System routes to Admin for approval. Admin reviews and approves. Status → APPROVED. Multi-level approval log recorded in po_approvals table. WhatsApp WA-01 to vendor: 'PO/2026-04/0018 approved. Amount ₹99,120. Delivery by [date]. View on portal.'

6 Vendor Accepts via Portal

VitroChem logs in to vendor portal. Views PO details. Clicks 'Accept'. Portal updates PO status → SENT. Priya's WhatsApp: 'Vendor VitroChem has accepted PO/2026-04/0018.'

7 Goods Arrive — GRN Created

3 days later, truck arrives with 195 kg (5 kg short-shipped). Ravi creates GRN against PO/2026-04/0018 in Warehouse module. Enters: Received 195 kg, Vehicle no GJ-01-AX-4521, LR no LR/VITRO/2026/0441. QC: 190 kg passed, 5 kg rejected (moisture issue — photo attached). GRN/2026-04/0019 created. Status: DRAFT.

8

Ravi Posts the GRN

Ravi clicks Post GRN. System: (a) Creates stock_ledger entry — GRN_IN, 190 kg, WH-MAIN, batch B-20260419. (b) Creates AP invoice for 190 kg × ₹420 = ₹79,800 + GST ₹14,364 = ₹94,164. Journal entry auto-posted: Dr Inventory ₹79,800, Dr GST Input ₹14,364 / Cr Accounts Payable ₹94,164. PDF GRN generated. WhatsApp WA-04 to Priya: 'GRN posted against PO/2026-04/0018. 190 kg accepted.'

9

3-Way Match

Mehul (Accountant) receives vendor invoice via email for ₹94,164. Enters invoice in system against GRN/2026-04/0019. System performs 3-way match: PO qty 200 → GRN qty 190 → Vendor invoice qty 190. Match successful. AP entry confirmed. Payment scheduled Net-30 per vendor terms.

10

Payment Processed

On day 30, Mehul clicks Pay on AP entry. Records NEFT payment ₹94,164. Bank UTR: SBIN0000001234567. AP cleared. Bank ledger updated. WhatsApp WA-08 to vendor: 'Payment of ₹94,164 sent. UTR: SBIN0000001234567.' PO status → RECEIVED (partial — 5 kg short). Debit note raised for 5 kg rejection.

Documents Generated in this Use Case:

Document	Reference No	PDF Generated?
Purchase Requisition	PR/2026-04/0023	No (internal record)
Purchase Order	PO/2026-04/0018	Yes — sent to vendor
Goods Receipt Note	GRN/2026-04/0019	Yes — internal QC record
Vendor AP Invoice	Auto-created	No (system record)
Payment Receipt	PAY/2026-04/0031	Yes — sent to vendor via WA

22.2 Use Case: Sales Order to Cash — Auto Parts Distributor

Company	Bharat Auto Components Ltd., Pune (Maharashtra — State 27)
Customer	Rajasthan Auto Traders, Jaipur (Rajasthan — State 08) ← inter-state, IGST applies
Actors	Deepak (Sales Manager), Anita (Warehouse), Mehul (Accountant)
Trigger	Customer WhatsApp: 'Need 500 pcs Brake Pads (SKU: FG-0078) urgently'

1

Deepak Creates Quotation

Deepak opens Sales → New Quotation. Selects customer Rajasthan Auto Traders (state 08). System detects: Seller state 27 ≠ Buyer state 08 → IGST applicable. Enters 500 pcs Brake Pad FG-0078 @ ₹185/pc. GST 18% IGST auto-applied. Subtotal ₹92,500 + IGST ₹16,650 = Total ₹1,09,150. QT/2026-04/0041 created. Quotation PDF generated and WhatsApp-shared to customer.

2

Customer Accepts — Convert to Sales Order

Customer replies 'Approved'. Deepak clicks 'Convert to Sales Order'. System runs credit limit check: Customer credit limit ₹2,00,000. Credit used ₹65,000. New order ₹1,09,150. Total ₹1,74,150 < ₹2,00,000. ✓ Pass. SO/2026-04/0038 created. 500 pcs of FG-0078 soft-reserved in WH-PUNE inventory. Status: CONFIRMED.

3

Anita Picks & Packs

Anita receives pick list notification. Navigates to Warehouse → Sales Orders → SO-0038. Scans barcode of Brake Pad boxes (batch B-20260301). Confirms 500 pcs picked. Packaging details: 10 boxes × 50 pcs. Weight: 25 kg. Delivery challan generated. Status → PROCESSING.

4

e-Way Bill Generated

Order value ₹1,09,150 > ₹50,000 AND inter-state → e-Way Bill mandatory. System calls NIC e-Way Bill API with transporter GJ02EX9999, distance 600 km. EWB No: 461234567890 returned and stored on Sales Order. Validity: 1 day (distance < 700 km by road).

5

Invoice Posted + IRN Generated

Deepak clicks Generate Invoice. System: (1) Creates GST invoice INV/2026-04/0051. (2) Calls IRP e-Invoice API. IRN (64-char hash) + QR code returned. (3) Invoice PDF generated with IRN, QR code, EWB number embedded. (4) AR entry created: Dr AR ₹1,09,150 / Cr Sales ₹92,500 / Cr IGST Payable ₹16,650. WhatsApp WA-05: Invoice PDF link sent to customer's WhatsApp.

6

Goods Dispatched

Truck departs. Anita marks Dispatched. Stock ledger: SALES_OUT -500 pcs FG-0078 WH-PUNE. Courier: BlueDart, docket BDL20260419001. SO status → DISPATCHED. Customer WhatsApp: 'Order dispatched. Track: BDL20260419001. Invoice attached.'

7

Payment Collected

Customer pays ₹1,09,150 via NEFT on due date. Mehul records payment in Finance → Payments. AR entry cleared. Bank ledger: Dr Bank ₹1,09,150 / Cr AR ₹1,09,150. WhatsApp WA-07 to customer: 'Payment of ₹1,09,150 received. Receipt attached. Thank you!'

22.3 Use Case: Production Run — Chemical Batch Manufacturing

Company	Chemico Industries, Vapi, Gujarat
Product	Sodium Hypochlorite Solution (FG-0112), BOM version v2.0 — produces 100 litres per batch
Actors	Nilesh (Production Supervisor), Admin
Trigger	Sales Order SO/2026-04/0038 confirmed but only 120 L in stock vs 200 L ordered

BOM v2.0 for Sodium Hypochlorite (100 L batch):

Component	Item Code	Qty per 100 L	UOM	Current Stock
Chlorine Gas	RM-0201	8.5	kg	45 kg
Caustic Soda Lye 48%	RM-0202	12.0	kg	80 kg
Demineralised Water	RM-0203	85.0	ltr	500 ltr
Stabilizer Additive	RM-0204	0.5	kg	10 kg

1

System Suggests Production

When SO/2026-04/0038 confirmed for 200 L and available stock = 120 L, system auto-suggests: 'Insufficient stock. Create Production Order for 80 L?' Nilesh clicks Yes. PRD/2026-04/0011 created with planned_qty = 80 L (rounded to 100 L for efficiency). System calculates material requirement: 8.5 kg × 1 = 8.5 kg Chlorine, 12 kg Caustic Soda, 85 L DM Water, 0.5 kg Stabilizer.

2 Material Availability Check

System checks stock: Chlorine 45 kg ✓, Caustic Soda 80 kg ✓, DM Water 500 L ✓, Stabilizer 10 kg ✓. All materials available. Status → RELEASED. Stock reservations created (soft-reserve) for all 4 raw materials. WhatsApp WA-09 to Nilesh: 'Production Order PRD/2026-04/0011 released. Batch: 100 L Sodium Hypochlorite.'

3 Nilesh Starts Production

Nilesh logs into the tablet on shop floor. Opens Production → PRD-0011. Clicks Start Production. actual_start timestamp recorded. Status → IN_PROGRESS. Production Order PDF printed with BOM checklist and QC checkpoints.

4 Material Consumption Logged

During production, Nilesh logs actual material consumption: Chlorine 8.3 kg (saved 0.2 kg), Caustic Soda 12.1 kg (0.1 kg extra), DM Water 85 L, Stabilizer 0.5 kg. Consumption saved in production_order_materials table.

5 Production Completed

Actual output: 98 L (2 L evaporation loss — within acceptable variance 3%). Nilesh clicks Complete. actual_qty = 98. actual_end timestamp recorded. Stock ledger: PRODUCTION_OUT for all RM consumed. PRODUCTION_IN for 98 L FG-0112 in WH-MAIN. Journal entry: Dr Finished Goods / Cr WIP / Cr Raw Material accounts. WhatsApp WA-10 to Admin: 'PRD/2026-04/0011 complete. Planned 100 L, Actual 98 L. Variance 2%.'

6 Sales Order Fulfilled

With 120 L existing + 98 L new production = 218 L available > 200 L ordered. SO/2026-04/0038 can now be fulfilled. Warehouse picks 200 L, processes dispatch normally.

22.4 Use Case: Monthly Payroll Processing

Company	Suncure Pharma Pvt. Ltd.
Pay Period	April 2026
Actors	Kavita (HR Manager), Mehul (Accountant), All Employees
Trigger	Last working day of April — Kavita initiates payroll run

1 Kavita Reviews Attendance

HR → Attendance → April 2026 summary. Reviews daily attendance records. Finds 3 employees with unapproved absences. Marks 2 as LOP (Loss of Pay), 1 approved as casual leave (pre-approved application). Attendance locked for payroll.

2 Leave Reconciliation

System reconciles approved leaves against leave balance. Casual leave balance deducted. LOP days calculated for each employee. HR confirms leave summary: 45 employees, total LOP days = 8 across 3 employees.

3 Kavita Initiates Payroll Run

HR → Payroll → New Run → April 2026. System pre-populates all active employees. payroll_runs record created. Status: CALCULATING.

4 Auto-Calculation

System computes for each employee: (1) Gross = Basic + HRA + Allowances – LOP deduction. (2) PF Employee = 12% of basic IF basic ≤ ₹15,000. (3) ESI Employee = 0.75% of gross IF gross ≤ ₹21,000. (4) Professional Tax per Maharashtra slab (₹200/month if gross > ₹10,000). (5) TDS as per income tax declaration. (6) Net = Gross – PF – ESI – PT – TDS – Other deductions. payroll_details rows created for each employee.

5 Kavita Reviews and Approves

HR reviews payroll summary: 45 employees, total net salary ₹18,42,300. Compares with last month — variance +2.1% (3 salary increments). Approves. Status → APPROVED.

6 Mehul Processes Bank Transfer

Finance → Payroll → April 2026 → Generate Payment File. NEFT/RTGS file in bank format (ICICI / HDFC compatible CSV) downloaded. Mehul uploads to corporate net banking. Confirms payment. All payroll_details records marked PAID. Journal entry posted: Dr Salary Expense ₹18,42,300 / Cr Bank ₹17,21,400 / Cr PF Payable ₹72,300 / Cr ESI Payable ₹14,800 / Cr PT Payable ₹9,000 / Cr TDS Payable ₹24,800.

7 Payslips Generated and Sent

System queues payslip PDF generation for all 45 employees. Each payslip generated with employee-specific data. WhatsApp WA-12 fired for each employee with their payslip as PDF attachment link. 'Dear [Name], your salary for April 2026 ₹[Net] has been credited to your account.' Payslips also available in employee self-service portal.

8 Statutory Filings

System generates: PF challan (ECR file for EPFO portal), ESI challan (ESIC portal format), PT challan for Maharashtra. Mehul downloads and files before 15th of next month.

Sample Payslip Calculation — Employee: Ramesh Patel (EMP-0023)

Component	Amount (₹)	Component	Amount (₹)
Basic Salary	14,000	PF Employee (12%)	1,680
HRA	5,600	ESI Employee (0.75%)	148
Conveyance	800	Professional Tax	200
Special Allowance	2,100	LOP Deduction (0 days)	0
GROSS EARNINGS	22,500	TOTAL DEDUCTIONS	2,028
		NET PAY	20,472

22.5 Use Case: GST Return Filing — Month-End

Accountant Mehul prepares and verifies GSTR-1 data for April 2026 at month-end.

1 Lock Period

Mehul navigates to Finance → GST → Lock Period → April 2026. System confirms no draft invoices pending for April. Period locked — no new invoices can be backdated.

2 GSTR-1 Data Verification

Finance → GST → GSTR-1 → April 2026. System aggregates: B2B invoices (32 invoices to GST-registered buyers), B2C invoices (8 invoices, unregistered), Credit notes (2 issued), Nil-rated supply (1 export invoice @0%). Mehul reviews each section. Flags 1 invoice with missing HSN — corrects it.

3 Export JSON for GST Portal

Clicks Export GSTR-1 JSON. System generates JSON in GSTN-specified schema. File downloaded: GSTR1_24AAICS1234P1Z5_042026.json. PDF summary report also generated for internal record.

4 GSTR-1 Filed

Mehul logs into GST portal, uploads JSON. Verifies data. Files GSTR-1 by 11th of May. ARN (Acknowledgement Reference Number) noted and saved in system settings.

5 GSTR-3B Preparation

Finance → GST → GSTR-3B → April 2026. System shows: Output Tax Liability ₹3,84,200 (from sales). ITC Available ₹1,47,600 (from purchases). Net Tax Payable = ₹3,84,200 – ₹1,47,600 = ₹2,36,600.

6 Tax Payment and GSTR-3B Filing

Mehul pays ₹2,36,600 via GST portal challan (NEFT). PMT-06 challan generated. GSTR-3B filed by 20th May. ARN saved. Finance record updated with GST payment journal entry.

22.6 Use Case: New Employee Onboarding

1 HR Creates Employee Record

Kavita navigates to HR → Employees → Add New. Enters: Name Suresh Mehta, DOB 12-03-1995, DOJ 21-04-2026, Department: Production, Designation: Operator, Employment Type: FULL_TIME. Basic Salary ₹13,500 (< ₹15,000 → PF applicable). PF Number: MH/ABC/1234567, ESI Number: 31-00-123456-000-0001. Auto-generated: EMP-0046.

2 User Account Created

Kavita checks 'Create Login'. System creates user account: suresh.mehta@suncure.erp with role 'Staff/Operator'. Password auto-generated, sent to employee's WhatsApp. Employee linked: employees.id ↔ users.employee_id.

3 Leave Balance Initialized

System auto-creates leave balance for FY 2026-27: Casual Leave 12 days, Sick Leave 7 days (pro-rated from April DOJ = 9 days CL, 5 days SL). Leave policy applied from department settings.

4 Attendance Enrollment

Biometric device sync (if integrated) or HR manually marks attendance from DOJ. First month partial — only days from 21st April counted.

5 First Payroll Inclusion

When May payroll runs, Suresh is automatically included. But April payroll was already processed on 30th April → Suresh gets April arrears processed in May's payroll (days 21-30 = 10 days). System calculates: $10/30 \times ₹13,500$ basic = ₹4,500 arrear added to May payslip with note 'April Arrear'.

22.7 Use Case: Warehouse Transfer & Stock Reconciliation

Inter-warehouse transfer of Brake Pads from WH-PUNE to WH-MUMBAI and subsequent reconciliation.

1 Transfer Request

Mumbai warehouse manager Anita creates Warehouse → Transfer → New. From: WH-PUNE. To: WH-MUMBAI. Item: FG-0078 Brake Pads. Qty: 200 pcs. Reason: Mumbai order fulfilment. Status: DRAFT.

2 Approval

WH-PUNE manager Ravi approves the outward transfer. Status → APPROVED.

3 Stock Deducted from Source

Ravi confirms dispatch. Stock ledger: TRANSFER_OUT –200 pcs WH-PUNE. Transfer challan PDF generated. Vehicle and LR number recorded.

4 Receiving at Destination

Anita receives goods at WH-MUMBAI. Confirms 198 pcs received (2 pcs damaged in transit). TRANSFER_IN +198 pcs WH-MUMBAI. Variance of 2 pcs flagged for insurance claim. Both ledger entries reference the same transfer document for traceability.

5 Monthly Stock Reconciliation

Admin runs Inventory → Reconciliation → WH-MUMBAI → April 2026. System computes: Opening stock + all GRN_IN + TRANSFER_IN – SALES_OUT – TRANSFER_OUT – PRODUCTION_OUT = Closing balance. Physical count entered. Variance report: 3 items with discrepancies. For each variance, system requires Adjustment Voucher with reason (damaged/theft/sampling/counting error). Adjustments posted as stock ledger ADJUSTMENT entries. Journal entries auto-posted.

22.8 Use Case: Vendor Portal Order Acceptance

VitroChem Suppliers receives a PO and interacts entirely through the vendor portal without any phone calls or emails.

1 Vendor Credentials Sent

When vendor is created with portal_enabled = true, WhatsApp WA-13 auto-sent to vendor.whatsapp: 'Your ManufactureERP portal access: URL: erp.suncure.com/vendor | Username: vitrochem@supplier.com | Password: [auto-generated]. Please change on first login.'

2 Vendor Logs In

VitroChem opens browser, goes to /vendor/login. Enters credentials. Forced password change on first login. Dashboard shows: 3 Open POs, 2 Pending Invoices, Payment status of last 5 transactions.

3 Vendor Reviews PO

Vendor clicks PO/2026-04/0018. Sees all line items, delivery date, shipping address, and payment terms. Downloads PO PDF. Clicks 'Accept PO'. System updates PO status → SENT. Priya (Purchase Manager) gets WhatsApp: 'VitroChem has accepted PO/2026-04/0018.'

4 Vendor Uploads Invoice

After delivering goods, VitroChem logs into portal. Goes to PO/2026-04/0018 → Upload Invoice. Enters: Vendor Invoice No VI-2026-0341, Invoice Date, Amount ₹94,164. Uploads scanned invoice PDF. System links to GRN/2026-04/0019. 3-way match auto-triggered in background.

5 Vendor Checks Payment Status

Vendor → Payments tab. Sees: INV VI-2026-0341 | Due: 19-May-2026 | Status: SCHEDULED. On payment day: Status → PAID | UTR: SBIN0000001234567. Vendor can download payment advice PDF.

End of SRS Addendum — Chapters 21 & 22

This document extends ManufactureERP SRS v1.0. The engineering team should treat these chapters as binding specifications for PDF template design, queue job implementation, and QA test case coverage. All use cases in Chapter 22 should have corresponding end-to-end integration tests in the test suite.